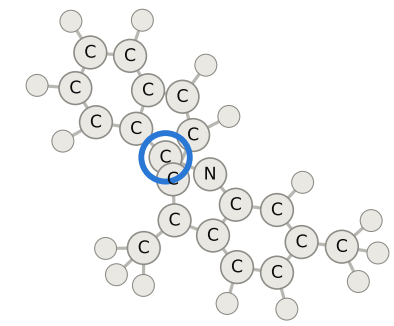
Most-cited prototypes. Units: L<layer>u<unit>; ¬ = must be 0; layer-0 units decoded into atom-count rules (n_X = number of X among the node and its neighbours)

P4 cited by 244 test explanations (mutagen 244)

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rule:  ¬L0u56 ∧ ¬L1u43 ∧ ¬L1u47 ∧ ¬L1u58 ∧ L2u4 ∧ L2u8 ∧ ¬L2u11 ∧ L2u21 ∧ L2u27 ∧ ¬L2u40 ∧ L2u49 ∧ L2u50 ∧ L2u59
training graphs with a fully matching node: mutagen 69%, nonmutagen 23%
cited as: some node is within 4 mismatches of P4  (x244)
   L0u56 = (n_H in [0.86, 3.00] ∧ n_I in [0.11, 2.00] ∧ n_NOT Cl in [1.00, 2.11] U [4.86, 5.00]) ∨ (n_C in [1.50, 5.00] ∧ n_H
       in [0.86, 3.00] ∧ n_I in [0.11, 2.00])
```

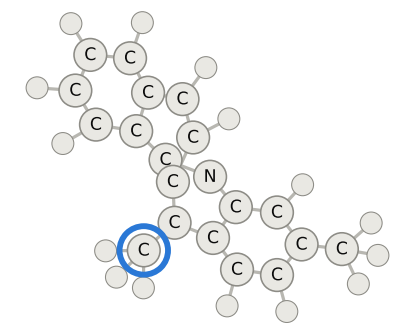
a best-matching training node (graph 0, mutagen)



P28 cited by 244 test explanations (mutagen 244)

```
rule:  ¬L0u49 ∧ ¬L1u6 ∧ ¬L1u17 ∧ ¬L1u33 ∧ ¬L1u48 ∧ L1u63 ∧ ¬L2u15 ∧ ¬L2u18 ∧ ¬L2u30 ∧ ¬L2u35
training graphs with a fully matching node: mutagen 97%, nonmutagen 95%
cited as: mean over nodes of match(P28) ≤ 0.73  (x244)
   L0u49 = (n_NOT C in [0.00, 0.86] U [4.46, 5.00] ∧ n_NOT Li in [2.42, 4.25]) ∨ (n_N in [0.75, 2.60] ∧ n_NOT C in [0.00,
       0.86] U [4.46, 5.00])
```

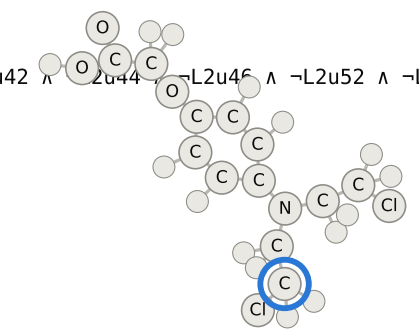
a best-matching training node (graph 0, mutagen)



P15 cited by 230 test explanations (mutagen 230)

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rule:  ¬L0u10 ∧ ¬L1u1 ∧ ¬L1u33 ∧ ¬L1u36 ∧ ¬L1u46 ∧ ¬L1u47 ∧ ¬L1u58 ∧ ¬L2u0 ∧ L2u1 ∧ ¬L2u7 ∧ ¬L2u11 ∧ L2u13 ∧ ¬L2u18 ∧ ¬L2u24 ∧ L2u27 ∧ ¬L2u41 ∧ L2u42 ∧ ¬L2u44 ∧ ¬L2u46 ∧ ¬L2u52 ∧ ¬L2u54 ∧ L2u60
training graphs with a fully matching node: mutagen 0%, nonmutagen 0%
cited as: mean over nodes of match(P15) in [0.00, 0.16] ∪ [0.31, 0.46] ∪ [0.62, 0.77] ∪ [0.92, 1.00]  (x230)
   L0u10 = (n_H in [0.86, 3.00] ∧ n_P in [0.24, 1.23] ∧ n_NOT H in [0.00, 2.00]) ∨ (n_C in [1.50, 5.00] ∧ n_H in [0.86, 3.00]
       ∧ n_P in [0.24, 1.23])
```

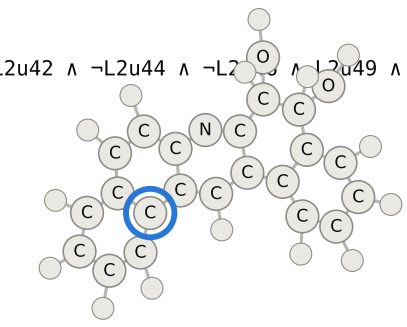
a best-matching training node (graph 11, mutagen)



P26 cited by 187 test explanations (nonmutagen 187)

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rule:  ¬L0u50 ∧ ¬L1u14 ∧ ¬L1u33 ∧ ¬L1u36 ∧ ¬L1u44 ∧ ¬L1u47 ∧ ¬L1u51 ∧ ¬L1u58 ∧ ¬L2u0 ∧ L2u4 ∧ ¬L2u6 ∧ ¬L2u7 ∧ ¬L2u11 ∧ ¬L2u15 ∧ ¬L2u34 ∧ ¬L2u35 ∧ L2u42 ∧ ¬L2u44 ∧ ¬L2u45 ∧ L2u49 ∧ ¬L2u57
training graphs with a fully matching node: mutagen 20%, nonmutagen 2%
cited as: mean over nodes of match(P26) ≤ 0.02  (x187)
   L0u50 = (n_P in [0.24, 1.23]) ∨ (n_H in [0.86, 3.00])
```

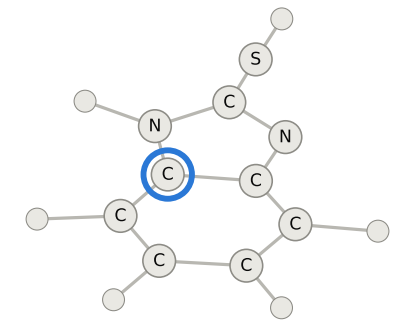
a best-matching training node (graph 230, nonmutagen)



P3 cited by 187 test explanations (nonmutagen 187)

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rule:  ¬L0u11 ∧ L0u24 ∧ ¬L0u31 ∧ ¬L0u36 ∧ ¬L1u44 ∧ ¬L1u58 ∧ ¬L2u4 ∧ ¬L2u15 ∧ ¬L2u40 ∧ ¬L2u57
training graphs with a fully matching node: mutagen 86%, nonmutagen 70%
cited as: mean over nodes of match(P3) ≤ 0.98  (x187)
   L0u11 = (n_C in [1.50, 5.00] ∧ n_O in [0.40, 1.41] U [2.41, 3.41] ∧ n_Cl in [0.56, 1.95]) ∨ (n_O in [0.40, 1.41] U [2.41,
       3.41] ∧ n_S in [0.41, 1.41] ∧ n_P in [0.24, 1.23])
   L0u24 = (n_NOT C in [0.00, 0.86] U [4.46, 5.00] ∧ n_NOT Li in [2.42, 4.25]) ∨ (n_N in [0.75, 2.60] ∧ n_NOT C in [0.00,
       0.86] U [4.46, 5.00])
   L0u31 = (n_P in [0.24, 1.23] ∧ n_NOT O in [0.00, 2.50]) ∨ (n_P in [0.24, 1.23] ∧ n_NOT H in [0.00, 2.00])
   L0u36 = (n_H in [0.86, 3.00] ∧ n_NOT P in [1.00, 3.12]) ∨ (n_H in [0.86, 3.00] ∧ n_NOT Cl in [1.00, 2.11] U [4.86, 5.00])
```

a best-matching training node (graph 15, nonmutagen)



P13 cited by 121 test explanations (mutagen 121)

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rule:  L0u24 ∧ ¬L1u39 ∧ ¬L1u58 ∧ ¬L1u61 ∧ ¬L2u7 ∧ L2u14 ∧ ¬L2u15 ∧ L2u17 ∧ ¬L2u35 ∧ ¬L2u51
training graphs with a fully matching node: mutagen 64%, nonmutagen 29%
cited as: mean over nodes of match(P13) in [0.14, 0.88]  (x121)
   L0u24 = (n_NOT C in [0.00, 0.86] U [4.46, 5.00] ∧ n_NOT Li in [2.42, 4.25]) ∨ (n_N in [0.75, 2.60] ∧ n_NOT C in [0.00,
       0.86] U [4.46, 5.00])
```

a best-matching training node (graph 0, mutagen)

